Interconnected

Towards the Orthogonal Technology Lab, v0.1

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These are notes towards setting up a research lab that doesn't yet exist. It doesn't *need* to exist; I'm learning by writing, and what I learn might lead anywhere (or nowhere).

In my Thingscon talk I ended with this call:

> *So I guess what I'm asking for is a different kind of think tank, not one that works with recommendations and reports and regulation, but* ***a new think tank that trades in politically opinionated, worked examples that demonstrate, demystify, and de-risk.***
>
> *– Matt Webb (Interconnected), The hard work of imagining, ThingsCon 2020*

What we need are visions of the future of technology that are values-driven, but "we don't need just design fictions. We need business model fictions, engineering feasibility study fictions, interop protocol specification fictions, investment return fictions."

My rationale is that it's *those* kind of worked examples that speak to the many groups that, together, shape the tech landscape. The purpose is to scout the path and shift the discourse.

This post is a first stab at defining a lab to invent and publish these worked examples. Do I believe this lab will happen? Well obviously it's something I would love to do but, no, not necessarily. In the spirit of Gedankenexperiment, the exercise itself is informative.

I'm not wedded to the name *"Orthogonal Technology Lab"* but I've chosen it as a placeholder from my post about orthogonal innovation, "a series of quite ordinary steps but simply in a different direction" - which can lead to a place very, very different from contemporary tech.

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The document version is **v0.1.** All I'm trying to do is capture and structure what's already in my head.

Research areas

Research areas TBD and there's a necessary mapping exercise, but I would want to cover areas such as:

- Connected products and zero data: e.g. how could you have voice controlled home lighting with no user sign-in and no cloud connection?
- Cloud services and open protocols for portability and interop: e.g. how share presence data between video call platforms, without insisting on federation?
- The consumerisation of Robotic Process Automation (RPA): e.g. could we ship a software-enabled co-op as a Shopify plugin?

What this *isn't* is about developing new tech for its own sake, or creating new business models where currently none are established. Which means no drones or cryptocurrency.

If there's an underlying thesis, it's to look in places where Srnicek's platform capitalism has taken hold, leading to data-driven captured marketplaces, and that's impeding technology development to the detriment of the consumer. Given that view, something like ad micro-targeting is a *symptom* of platform capitalism and not something to be rethought directly. Instead we start by rethinking the captured activity itself.

So there's a focus on B2C, rather than enterprise, and there's a focus on the smart assembling of existing tech rather than innovating new tech - though that's not to say that patentable technology won't be found.

Outcomes

The desired outcome is influence in shipped products and services, and that's tough to quantify, especially as an independent lab.

There are a couple of models here:

- As a think tank, what's the press coverage: articles with readership, video views, social media mentions and so on.

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- As an academic research lab, like a micro DeepMind, how many papers are published, and how much are they cited.
- As a commercial research lab, how many patents are granted - and, in the long run, what is the patent license income.

None of these are particularly good proxies to influence, except perhaps patent licensing.

Outputs

Artefacts:

The idea is to follow early product innovation processes, but ship all the collateral *around* the product rather than the product itself. So...

- a proof of concept of a zero-data connected product platform - but mainly the strategy deck that shows why this is a sound business.
- a proof of concept of interoperating video platforms - but mainly the open protocol that makes it possible, together with docs and protocols.
- a proof of concept of how mutualised businesses can run as software - and a shipped use case of it plugging into a gig economy platform.

Engagement:

Innovation doesn't happen in a vacuum. So communications and audience engagement are an output all of their own - everything from continuously explaining the work, to open research newsletters. Although these aren't the primary outcomes, it feels important to translate work into policy papers and MBA decks too.

There are a couple of non-traditional activities that are worth looking at here:

- A fellowship program, as discussed by Peter Bihr.
- An original art programme, as discussed (and used) by the Open Data Institute.

Commercial opportunities:

Eventually you'd want a lab like this to stand up on its own.

- Does that mean patents and licensing, in the style of Fraunhofer and mp3?
- Is there a membership model for early access, paid by one or more corporate sponsors?
- Is there a consultancy arm?
- Or is there an attached startup studio, with payback from equity?

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The trick is to pick a model with aligns with the mission (without adding too much overhead), and focus on that from day 1. It may not come to fruition, but I think it's necessary as a hedge against the loss of funding.

Structure

Process:

Putting aside the engagement and commercial activities, the core activity probably follows a standard innovation pipeline. There's concepting, prototyping, and development work, all separated by gates, and a healthy cull at each gate.

Portfolio management is about ensuring all stages of the pipeline are active. Looking at the four types of innovation I would say this is more about "disruptive" and "routine" innovation, where there are new business models but no new fundamental tech. But the outcomes may point the way to future tech research.

Team:

One question is about how much is done internally, within the lab, versus how much is done by commissioning.

My first best guess that there's a small, multi-disciplinary internal team of design, tech architecture, and business analysts, and that's the early, concepting part of the timeline.

Engagement/communications and program management are also in-house.

Then prototypes, technical protocols, and other builds are all commissioned.

You probably also need a permanent fundraising or partnerships function, plus management. Other support functions such as owning the art and fellowship programs, and commercialisation/patents, can probably be done on a freelance basis.

Talking about the team is a long way downstream from what matters - the outcomes - but it's worth sketching out. What is this… 8-10 people plus a strong commissioning budget?

Funding

The purpose of the above is to figure out the funding requirements.

What's a full innovation cycle… Three years? Four years? What's the full loaded budget for that, given the team and activities? It should just be a matter of doing the sums to come up with the number.

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That will give an indication of whether funding the lab is possible with grant funding, or whether there needs to be a different approach.

Questions and next steps

These are v0.1 thoughts, with the main purpose being to get them out of my head.

I'm not ready to get my pitch deck and spreadsheet out quite yet. Like I said, this is a thought experiment, and working it through will teach me something. Maybe I can take what I learn into a client engagements in adjacent areas, or maybe it'll give me a lens to understand historic research labs.

But here are the specifics that I'm thinking through, as a result of writing the above:

- **How do I better define the research areas?** Are they too narrow or too broad? How do I know what is out of scope? (This interacts with the funding model.)
- **Are there established models for the process?** For example, Startup Studio Playbook (Attila Szigeti) is fantastic for the macro, micro, and case studies of startup studios - and would be the perfect springboard if you were setting up a new one. But for research labs and think tanks?
- **Does the team make sense?** Especially the use of mixed internal/external R&D?
- Ultimately: **Feasibility.** Is there room and interest in the world for the Orthogonal Technology Lab?

Next steps: Socialising these ideas (hence this blog post), listening to feedback, and seeing whether thoughts are sparked, directly or indirectly.

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